Customer Shopping Behavior Analysis

1. Project Overview

The Customer Shopping Behavior Analysis project focuses on understanding customer purchasing patterns and shopping preferences through data analysis. The project uses a dataset containing 3,900 customer transactions across different product categories.

The main objective is to analyze customer spending habits, identify purchasing trends, understand customer segments, and evaluate the impact of discounts and subscriptions. The insights gained from this analysis can help businesses improve customer engagement, optimize marketing strategies, and make better business decisions.

2. Dataset Summary

The dataset contains customer-related information and transaction details that help in understanding shopping behavior.

- **Total Rows:** 3,900
- **Total Columns:** 18
- **Key Features:**
  - Customer Information: Age, Gender, Location, and Subscription Status.
  - Purchase Information: Items Purchased, Product Categories, Purchase Amount, Season, Size, and Color.
  - Shopping Preferences: Discount Applied, Promo Code Used, Purchase Frequency, and Previous Purchases.
  - Customer Feedback: Review Ratings and Shipping Type.
- **Missing Values:** 37 missing entries were identified in the Review Rating column.

The dataset provides a useful foundation for exploring customer preferences, spending patterns, and purchasing trends

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3. Exploratory Data Analysis using Python

Python was used for data exploration, cleaning, and preparation to ensure the dataset was suitable for further analysis.The following steps were performed:

- **Data Loading:** Imported the dataset into Python using the Pandas library.

Initial Data Inspection: Used functions such as `info()` and `describe()` to understand the dataset structure, data types, and statistical summaries.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- **Missing Value Treatment:** Identified missing values in the Review Rating column and replaced them using the median rating of the respective product categories.
- **Column Standardization:** Converted column names into snake_case format to maintain consistency and improve readability.
- **Feature Engineering:**
  - Created an `age_group` column to categorize customers based on their age.
  - Generated a `purchase_frequency_days` column to represent customer purchasing frequency.
- **Data Consistency:** Examined the relationship between `discount_applied` and `promo_code_used` and removed the redundant `promo_code_used` column.
- **Database Integration:** Established a connection between Python and PostgreSQL and transferred the cleaned DataFrame into the database for further SQL analysis.

4. Data Analysis using SQL (Business Transactions)

PostgreSQL was used to perform business-oriented queries and extract meaningful insights from customer transaction data.
The following analytical questions were explored:

1. **Revenue by Gender:** Analyzed the total revenue generated by male and female customers to understand differences in purchasing contributions.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users:** Identified customers who purchased products using discounts but still spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 839 | Query complete 00:00:00.350

3. **Top 5 Products by Rating:** Evaluated customer feedback to identify the five products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison:** Compared the average purchase amounts of customers choosing Standard and Express shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers:** Examined differences in average spending and total revenue between subscribed and non-subscribed customers.

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645
2	No	2847	59.87	170436

6. **Discount-Dependent Products:** Identified five products with the highest percentage of discounted purchases to understand their dependency on promotional offers.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation:** Divided customers into three categories — New, Returning, and Loyal — based on their purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category:** Identified the three most frequently purchased products within each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145

Total rows: 11 | Query complete 00:00:00.400

9. **Repeat Buyers & Subscriptions:** Investigated whether customers with more than five purchases showed a higher likelihood of subscribing.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group:** Calculated the total revenue contributed by different age groups to identify important customer segments.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

These SQL queries helped transform raw transaction records into meaningful business insights.

5. Dashboard in Power BI

Power BI is used to create an interactive dashboard that presents the results of the customer shopping behavior analysis in a clear and visually appealing format.
The dashboard provides a consolidated view of customer transactions, purchasing trends, and business performance.
It helps users explore different aspects of shopping behavior through visual representations, making it easier to identify patterns, compare customer segments, and understand important business metrics.
The dashboard makes the analysis more accessible and supports data-driven decision-making.



6. Business Recommendations

Based on the analysis, the following business recommendations can help improve customer engagement and overall business performance:

- **Boost Subscriptions:** Introduce exclusive offers, rewards, and additional benefits to encourage more customers to subscribe.
- **Customer Loyalty Programs:** Develop reward programs for repeat customers to strengthen customer relationships and increase long-term retention.
- **Review Discount Policy:** Analyze the effectiveness of discounts and promotional offers to balance customer attraction with business profitability.
- **Product Positioning:** Give more visibility to highly rated and frequently purchased products through promotions and marketing campaigns.
- **Targeted Marketing:** Focus marketing activities on high-revenue age groups and customers who prefer Express shipping to improve campaign effectiveness.